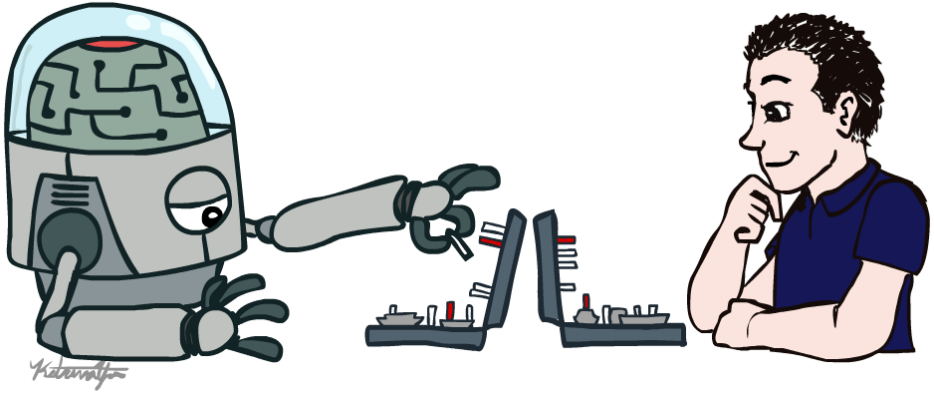
Artificial Intelligence

Introduction



Yan Lyu

About me

吕妍，博导，副教授

- 2016年获香港城市大学博士学位
- 2017年至2020年任新加坡国立大学博士后研究员
- 2020年10月起在东南大学计算机科学与工程学院任职副教授

研究方向：

- 大数据分析及可视化、时空数据挖掘
- 智慧城市、城市计算
- 智能交通
- 基于位置的社交网络、推荐系统

学院个人主页

My Research: Spatial Temporal Data

My Research: Spatial Temporal Data

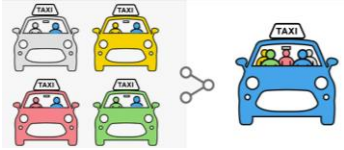
Data Visualization

Understanding Movements of Individuals and Crowds

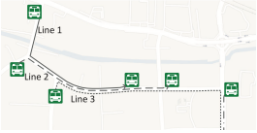
Recommendation

Intelligent Transportation

Taxi Sharing



Bus Line Planning



Autonomous Navigation

My Research: Spatial Temporal Data

Imitation Learning

Textbook

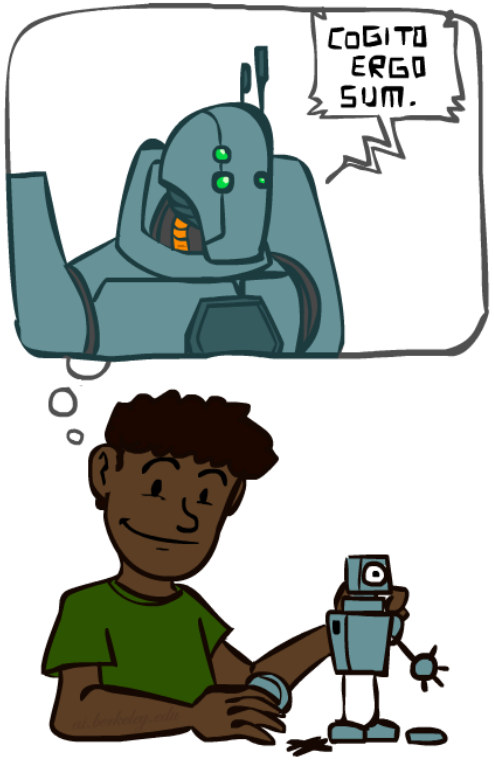
1. **[主教材]** 阿斯顿·张, 李沐 等. **《动手学深度学习》（PyTorch版）**. 人民邮电出版社, 2023. (配套在线资源: https://zh.d2l.ai/)
2. **[参考书-大模型]** Sebastian Raschka. **《Build a Large Language Model (From Scratch)》**. Manning Publications, 2024.
3. **[参考书-强化学习]** Richard S. Sutton. **《强化学习（第2版）》**. 电子工业出版社, 2019.
4. **[前沿技术文档]**
    1. Dan Jurafsky. *Speech and Language Processing (3rd ed. draft)*. (针对LLM理论)
    2. *Hugging Face / OpenAI / LangChain 官方技术文档*. (针对RAG与Agent开发实验)

Course Information

- 作业:
  - Project 1：经典机器学习：端到端的数据挖掘与预测 （25%，1-4周）
  - Project 2：深度学习与视觉：基于 PyTorch 的图像识别 （25%，5-8周）
  - Project 3：强化学习：训练你的专属游戏 AI Agent （25%，8-12周）
  - Project 4：生成式 AI 时代：构建本地私有化 RAG 智能问答助手 （25%，12-16周）
- 考核:
  - 期末汇报：随机抽取四个项目中任意一个项目进行期末汇报。

Today

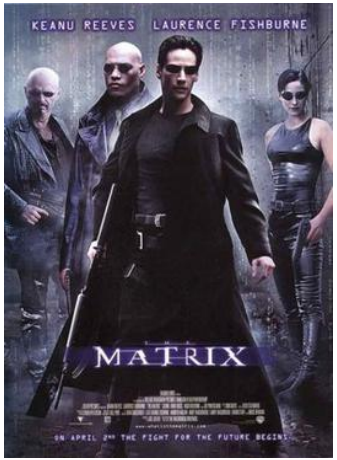
- What is artificial intelligence?
- What can AI do?
- What is this course?



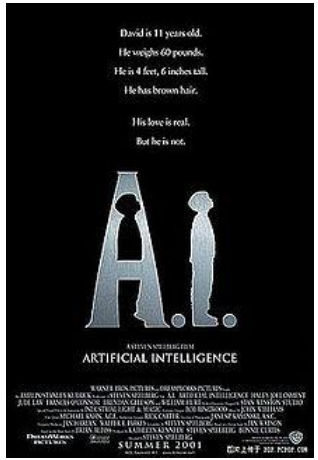
What is AI?

AI in fiction movies:

1984



1999



2001

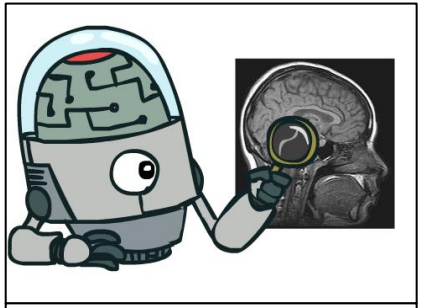
2008

2014

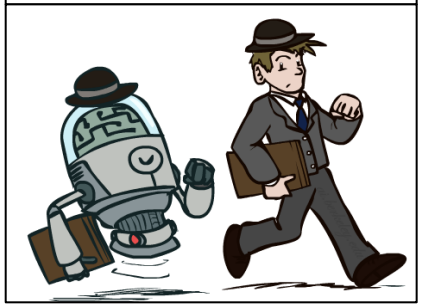
What is AI?

The science of making machines that:

Think like people

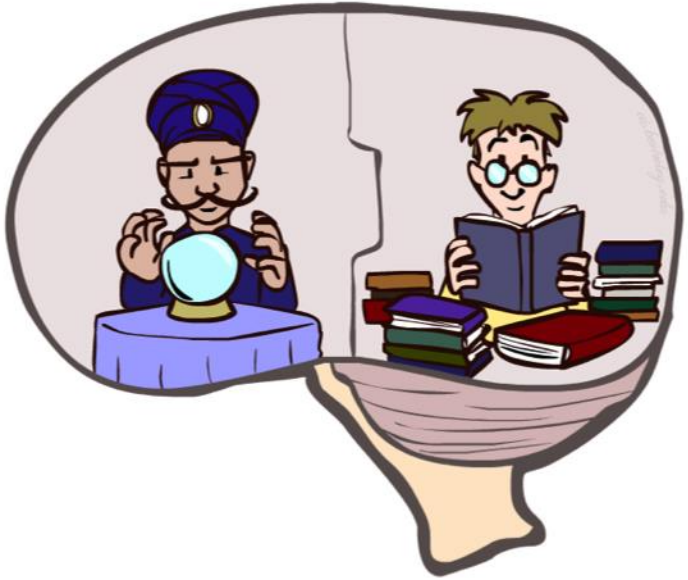


Act like people



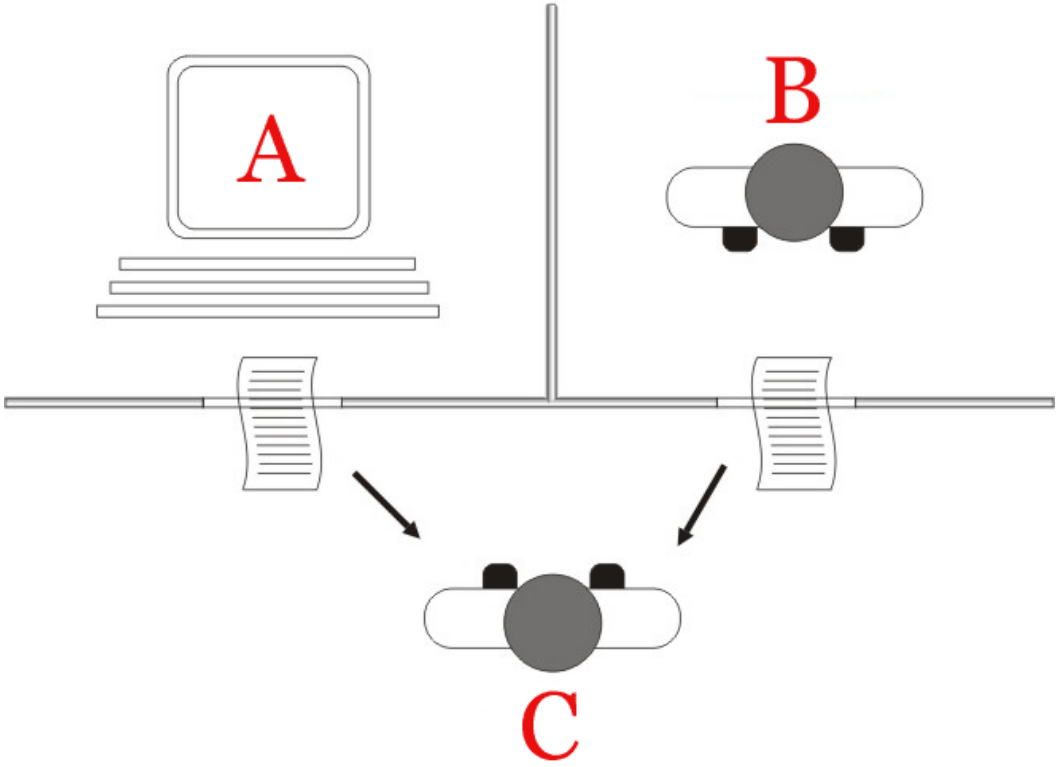
What About the Brain?

- Brains (human minds) are very good at making rational decisions, but not perfect
- Brains aren't as modular as software, so hard to reverse engineer!
- "Brains are to intelligence as wings are to flight"
- Lessons learned from the brain: **memory** and **simulation** are key to decision making



https://longtermrisk.org/birds-brains-planes-and-ai-against-appeals-to-the-complexity-mysteriousness-efficiency-of-the-brain/

Acting like human: The Turing Test

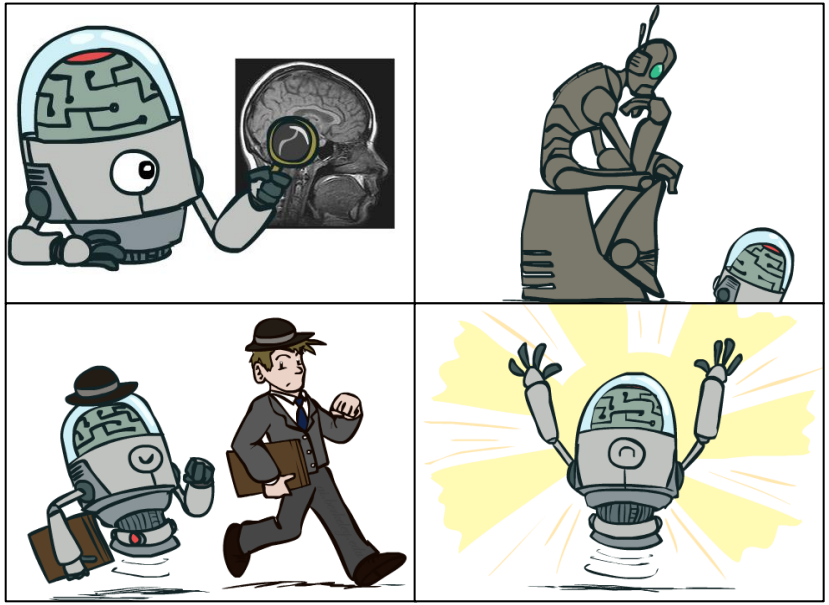


What is AI?

The science of making machines that:

Think like people

Think rationally



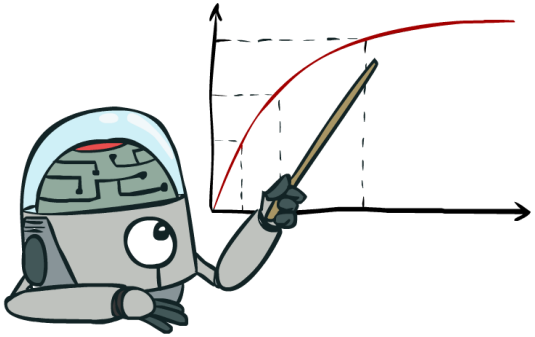
Act like people

Act rationally

Rational Decisions

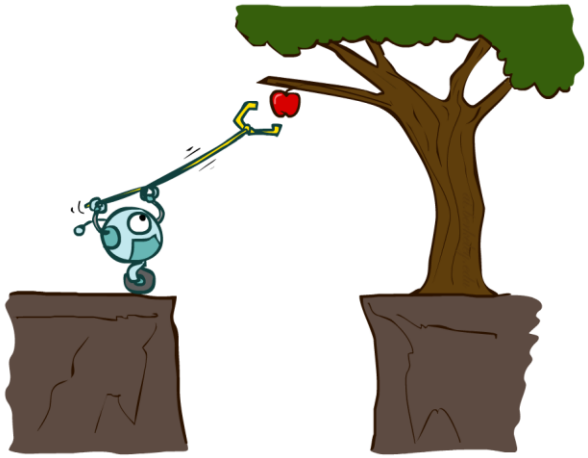
We'll use the term **rational** in a very specific, technical way:

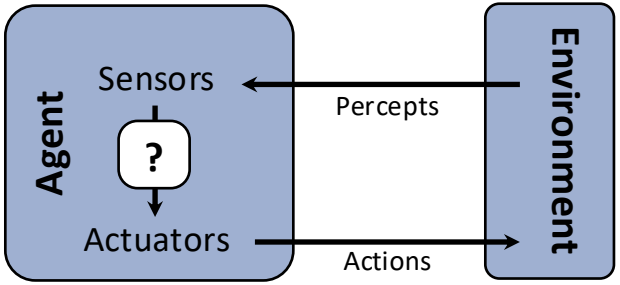
- Rational: maximally achieving pre-defined **goals**
- Rationality only concerns what decisions are made (not the thought process behind them)
- **Goals** are expressed in terms of the **utility** of outcomes
- Being rational means **maximizing your expected utility**



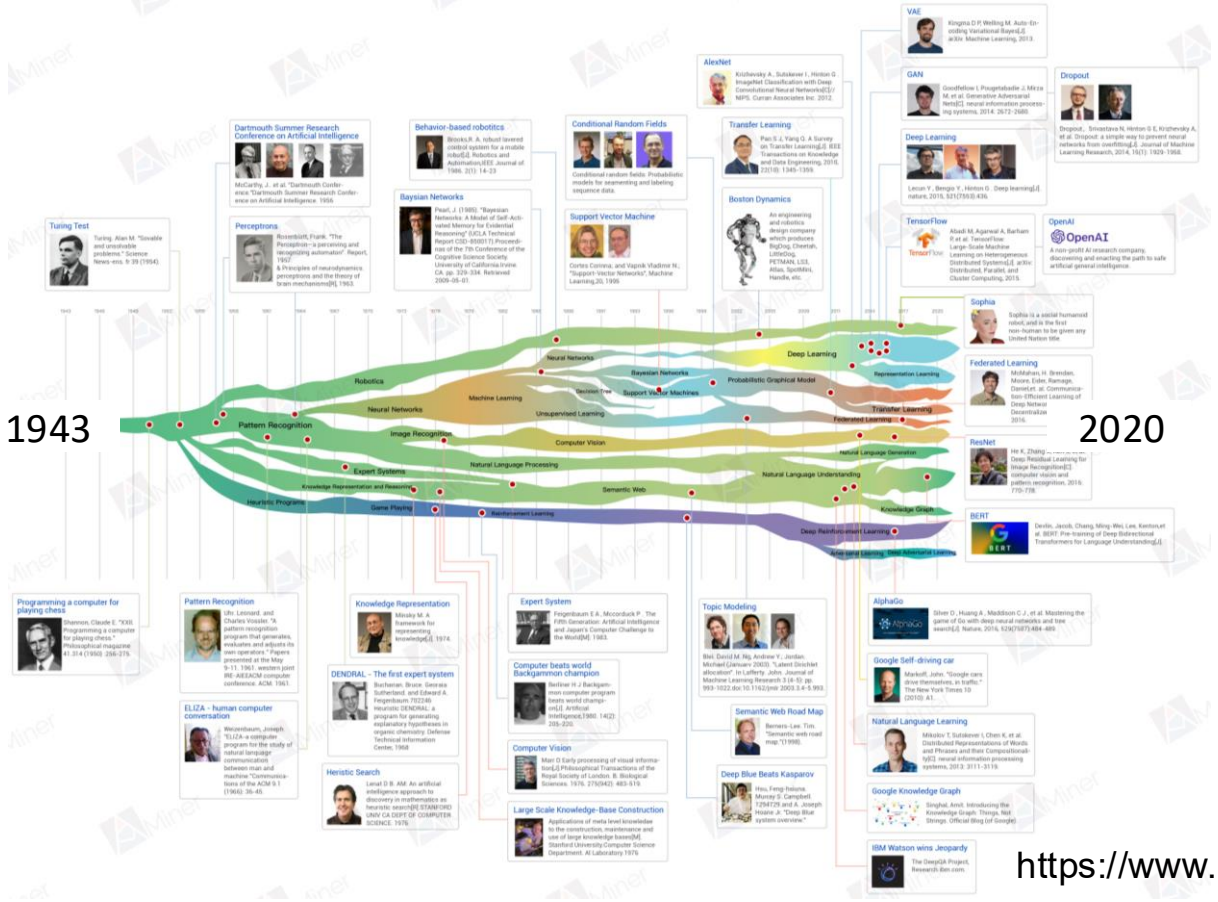
Designing Rational Agents

- An **agent** is an entity that *perceives* and *acts*.
- A **rational agent** selects actions that maximize its (expected) **utility**.
- Characteristics of the **percepts, environment,** and **action space** dictate techniques for selecting rational actions



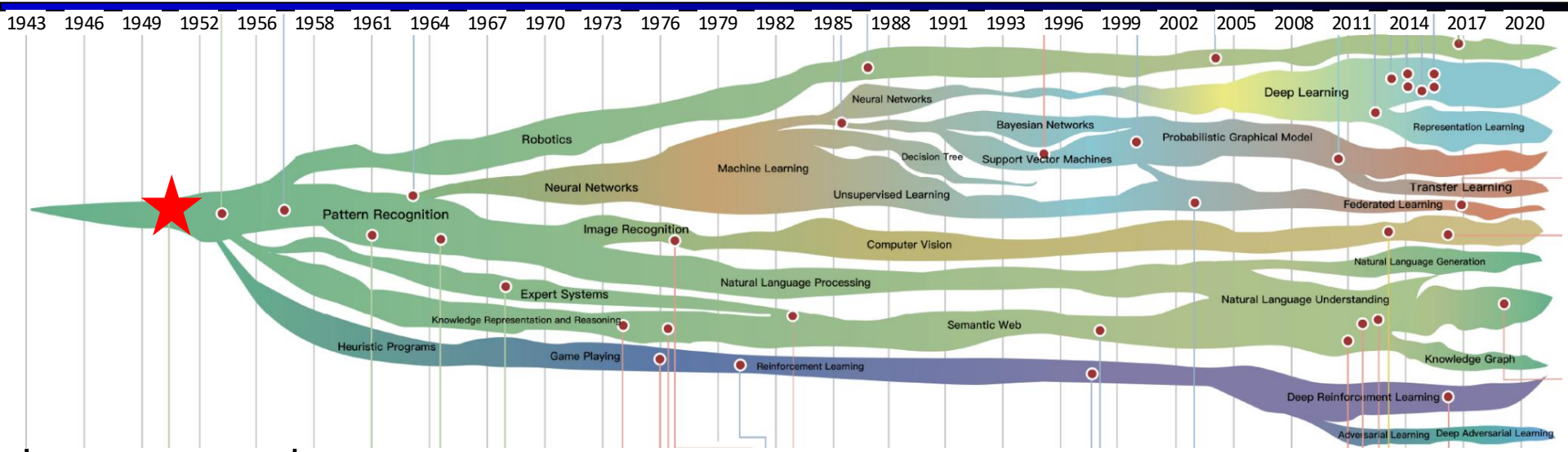


A (Short) History of AI



https://www.aminer.cn/ai-history

A (Short) History of AI

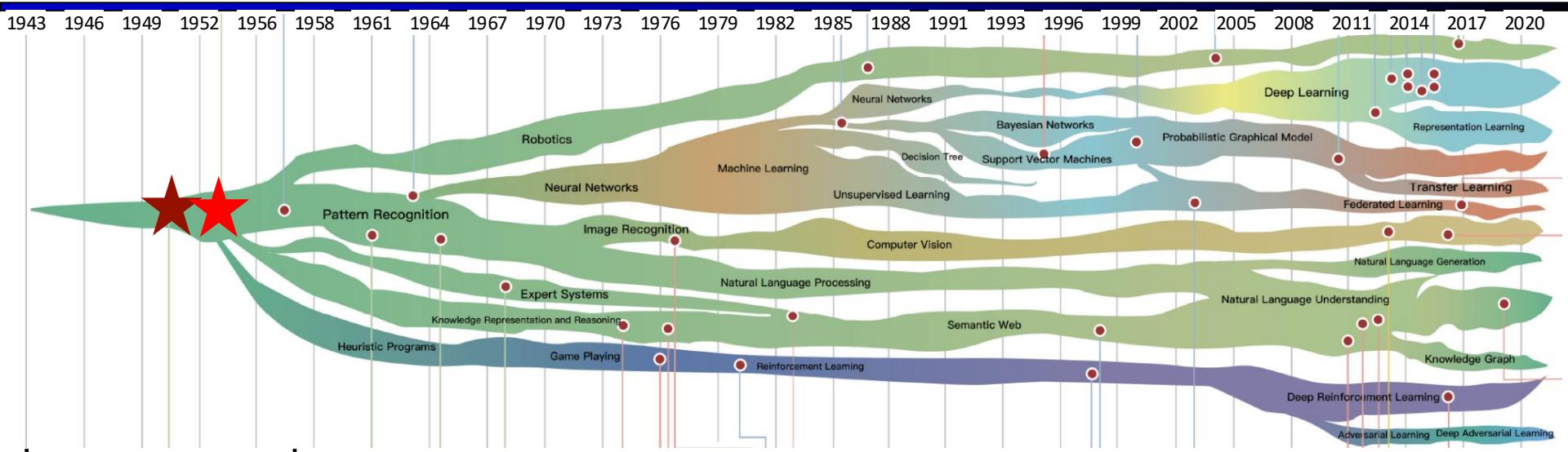


The birth

Programming a computer for playing chess

Shannon, Claude E. "XXII. Programming a computer for playing chess." Philosophical magazine 41.314 (1950) :256-275.

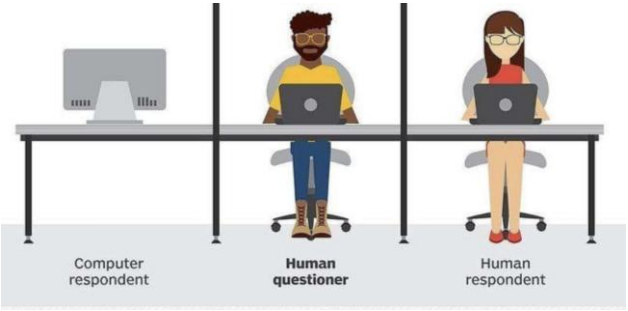
A (Short) History of AI



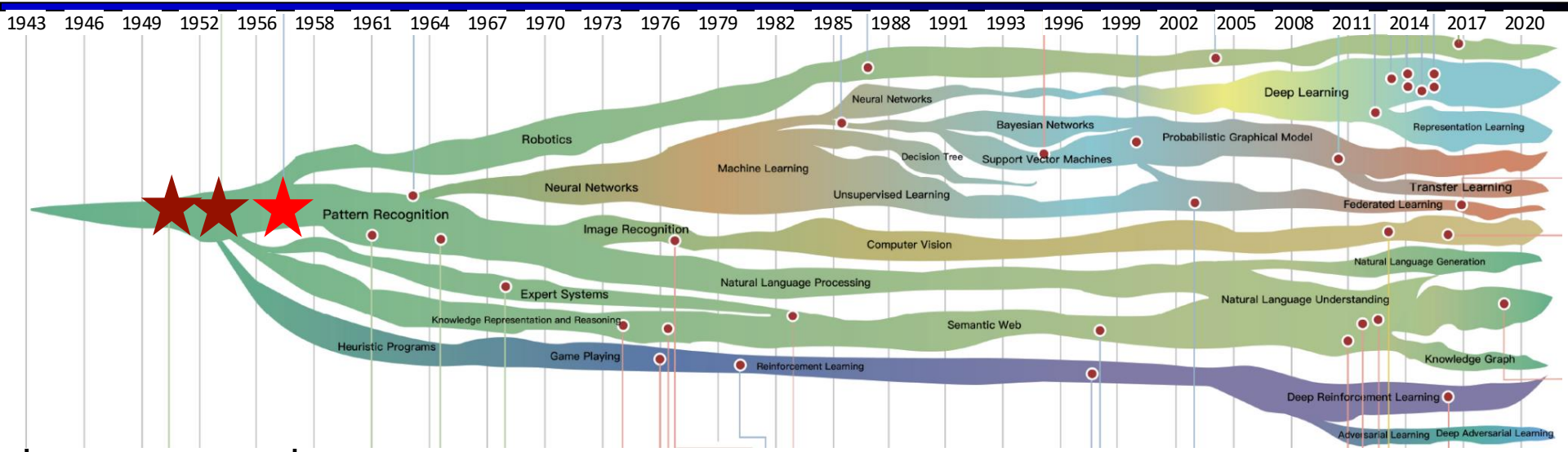
The birth

Turing Test

Turina. Alan M. "Sovable and unsolvable problems." Science News-ens. fr 39 (1954).



A (Short) History of AI

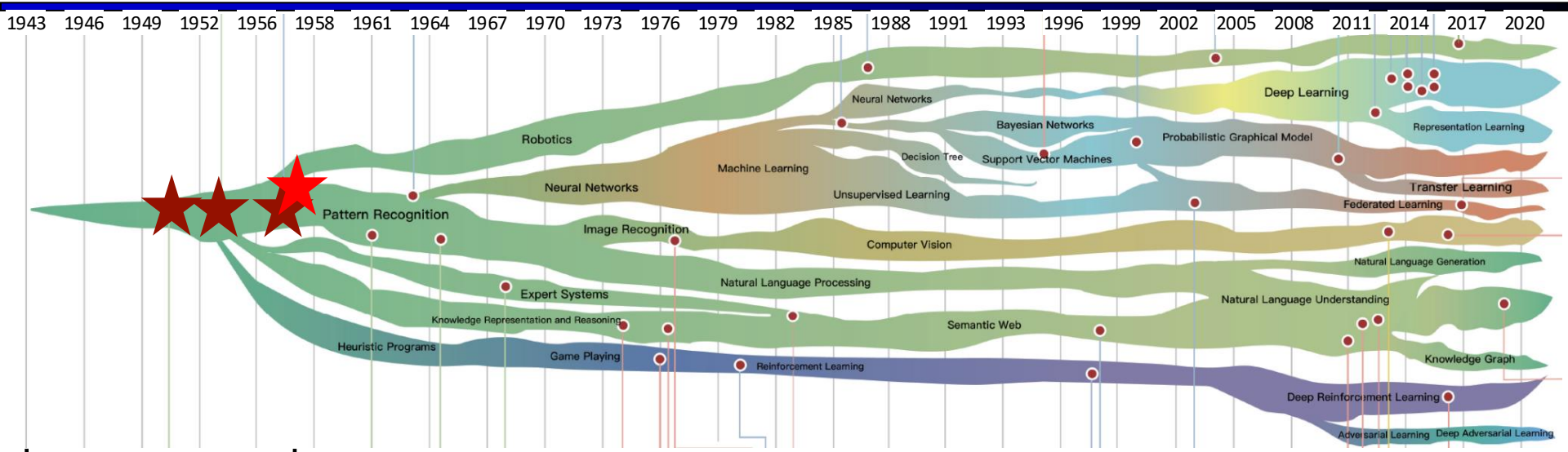


The birth

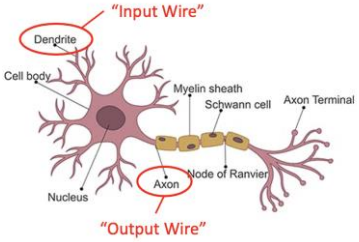
Dartmouth Summer Research Conference on Artificial Intelligence

McCarthy, J.. et al. "Dartmouth Conference."Dartmouth Summer Research Conference on Artificial Intelligence. 1956

A (Short) History of AI

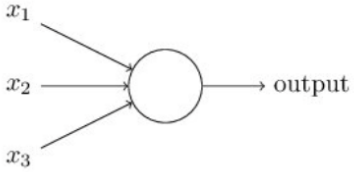


The birth



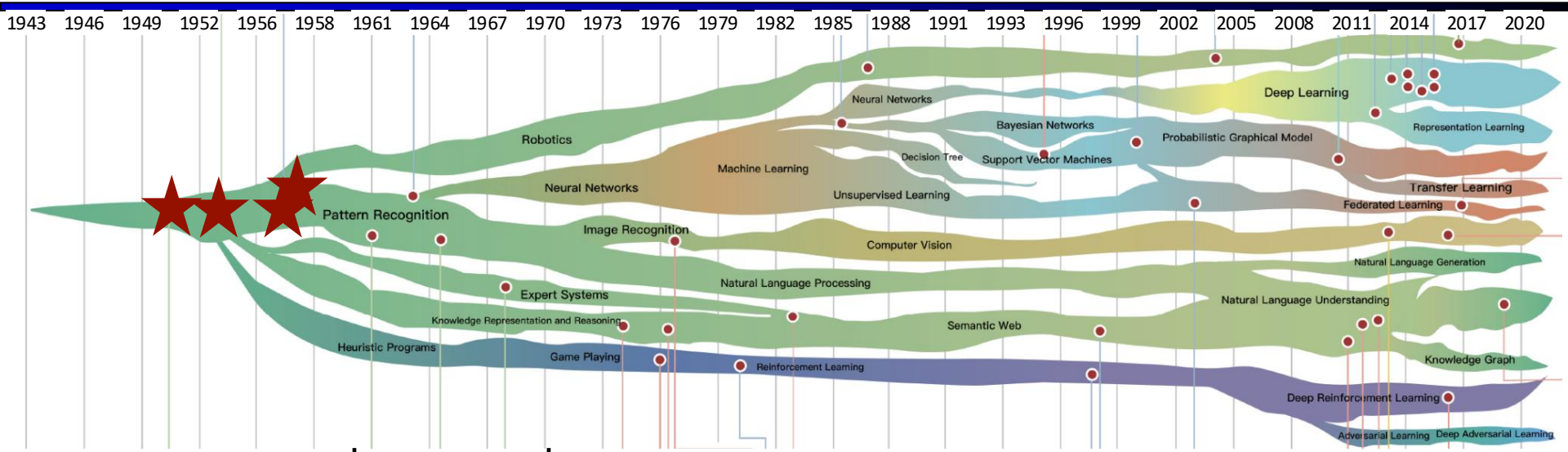
Perceptrons

Rosenblatt F. Principles of neurodynamics. Perceptrons and the theory of brain mechanisms[J]. American Journal of Psychology, 1963, 76(4).



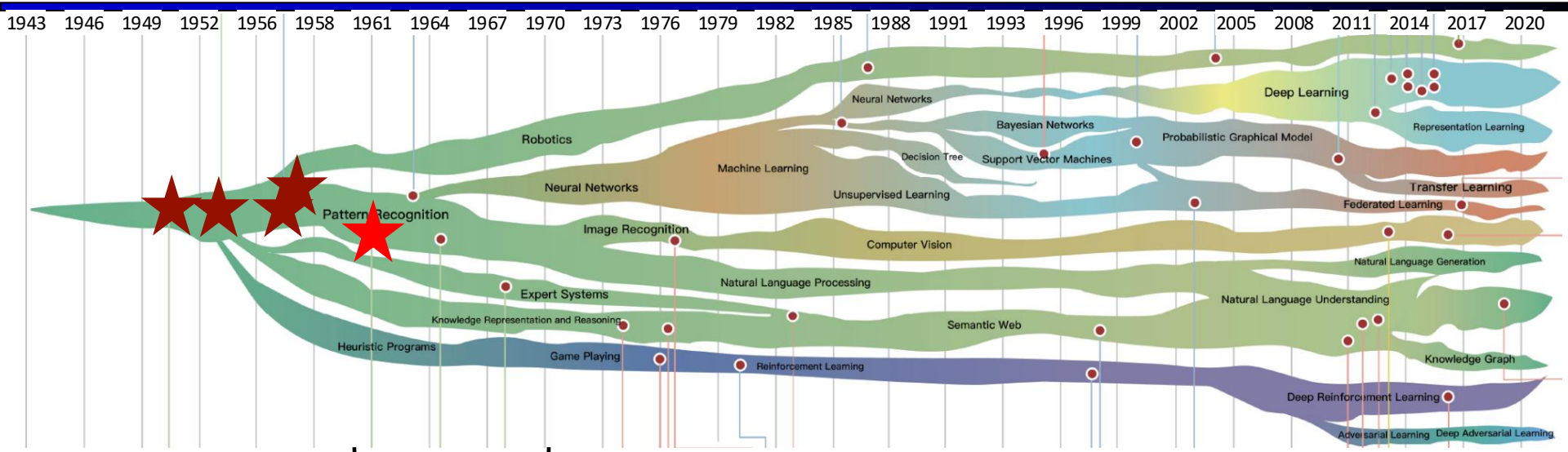
$$\text{output} = \begin{cases} 0 & \text{if } \sum_j w_j x_j \leq \text{ threshold} \\ 1 & \text{if } \sum_j w_j x_j > \text{ threshold} \end{cases}$$

A (Short) History of AI



The First Wave

A (Short) History of AI



The First Wave

Pattern Recognition

Uhr. Leonard. and Charles Vossler. "A pattern recognition program that generates, evaluates.and adjusts its own operators." Papers presented at the May 9-11. 1961. western joint IRE-AIEEACM computer conference. ACM. 1961.

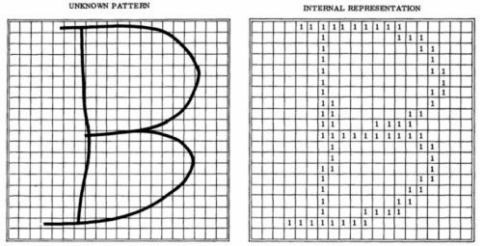
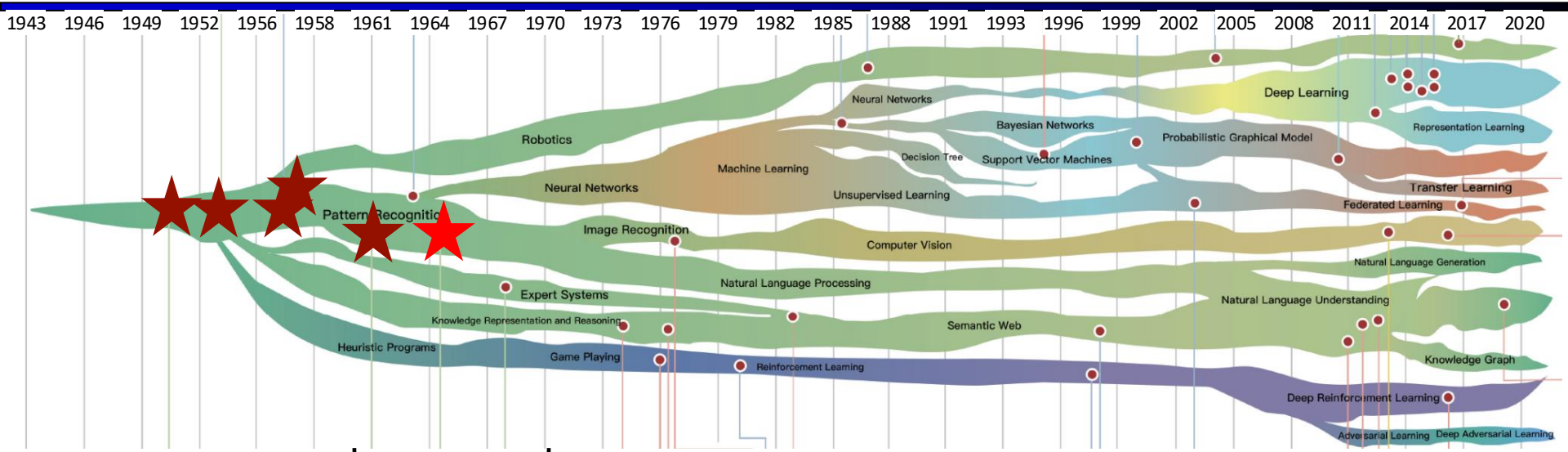


Figure 1 An unknown pattern is input as a 20 x 20 matrix with the cells covered by the pattern represented by '1's' and the other cells by '0's.'

A (Short) History of AI

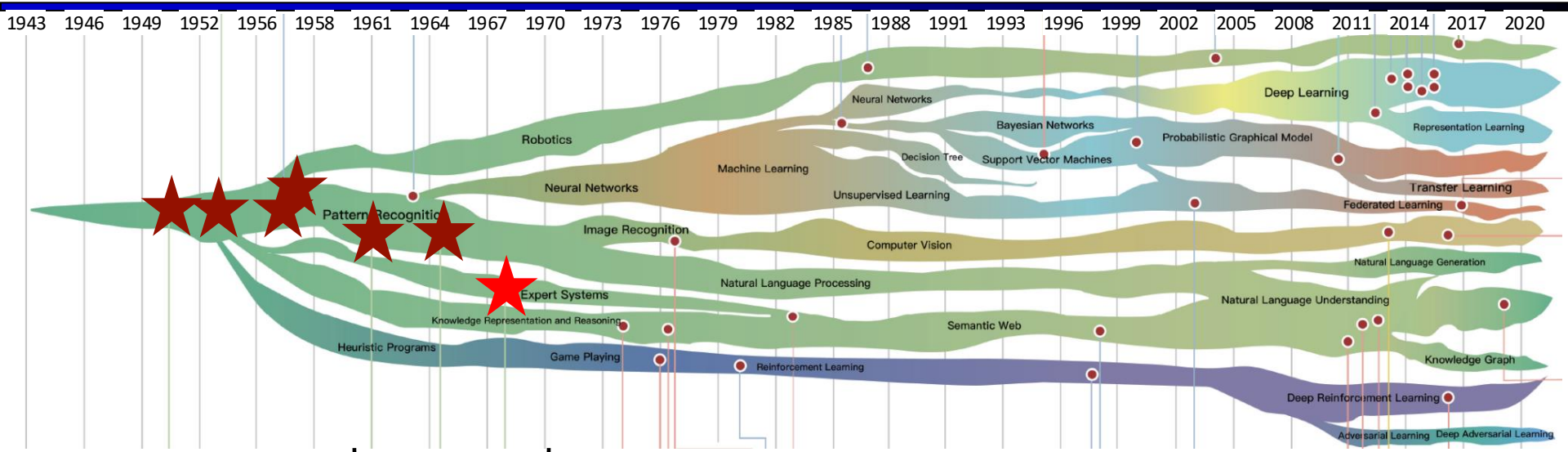


The First Wave

ELIZA - human computer conversation

Weizenbaum, Joseph. "ELIZA-a computer program for the study of natural lanquage communication between man and machine."Communica-tions of the ACM 9.1 (1966): 36-45.

A (Short) History of AI

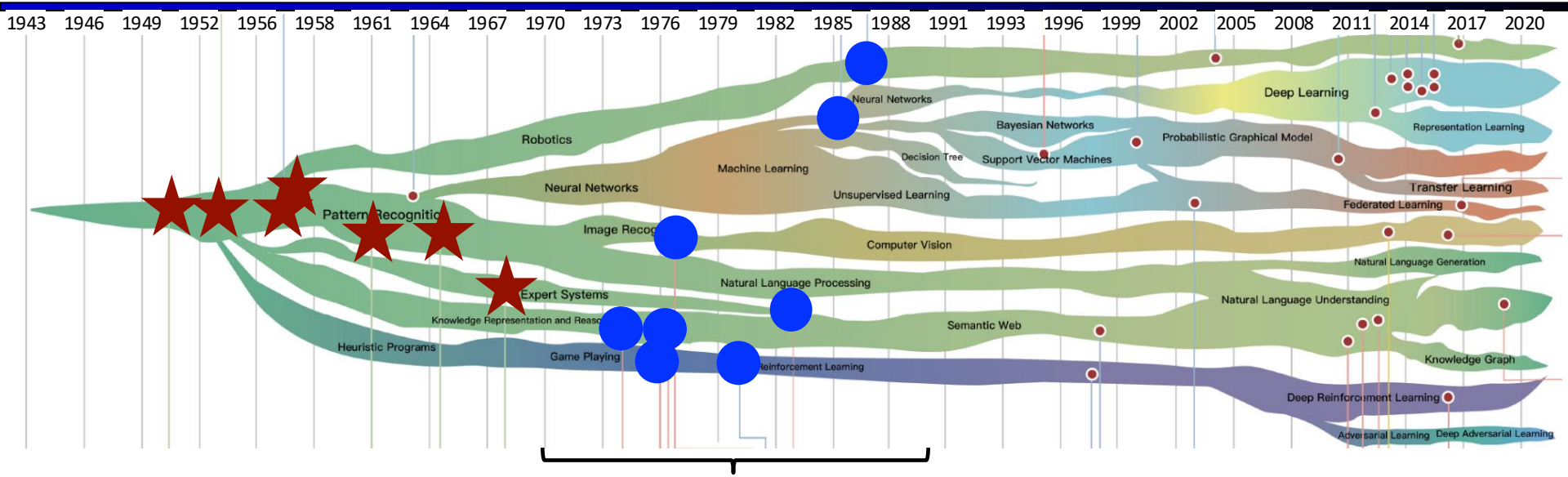


The First Wave

DENDRAL - The first expert system

Buchanan. Bruce. Georaia Sutherland. and Edward A. Feigenbaum.702246 Heuristic DENDRAL: a program for generating explanatory hypotheses in organic chemistry. Defense Technical Information Center, 1968

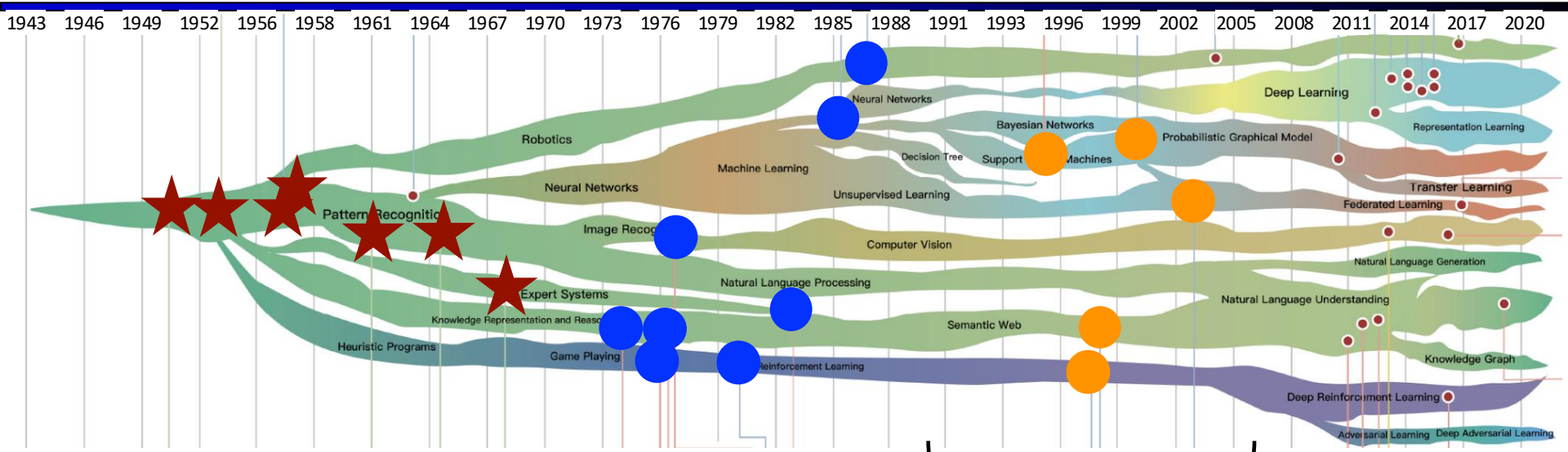
A (Short) History of AI



The Second Wave

- Knowledge Representation
- Heuristic Search
- Large Scale Knowledge-Base Construction
- Computer Vision
- Expert System
- Computer beats world Backgammon champion
- Baysian Network
- Behavior-based Robotics

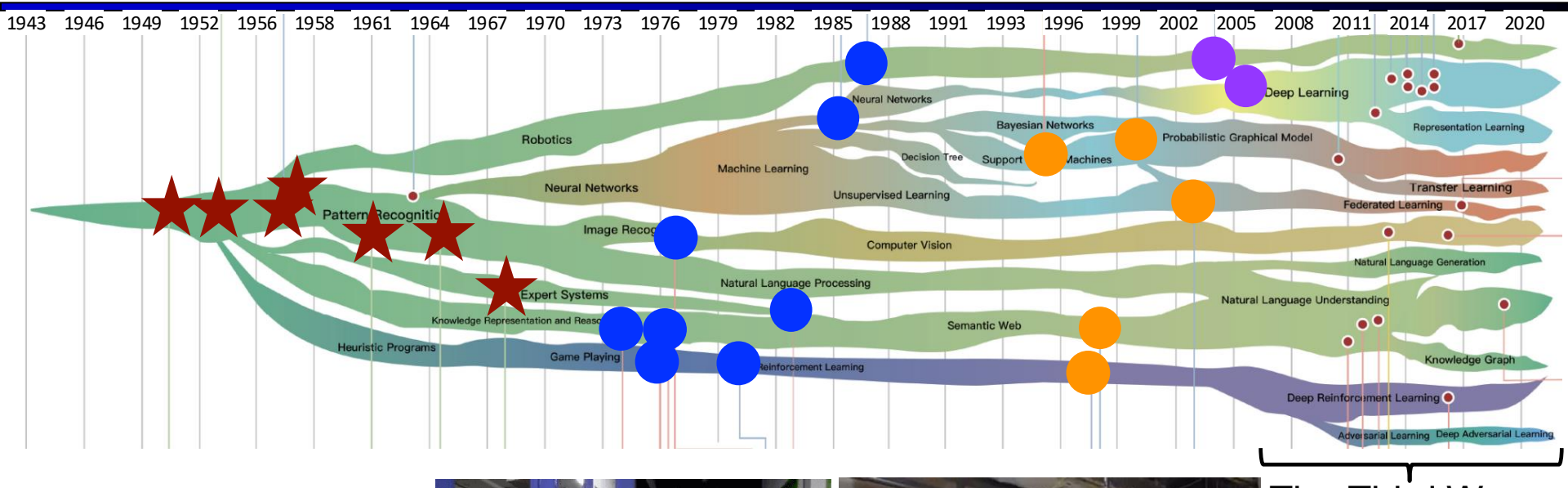
A (Short) History of AI



Steady Development

- Support Vector Machine
- Conditional Random Fields
- Topic Modeling
- Semantic Web Road Map
- Deep Blue Beats Kasparov

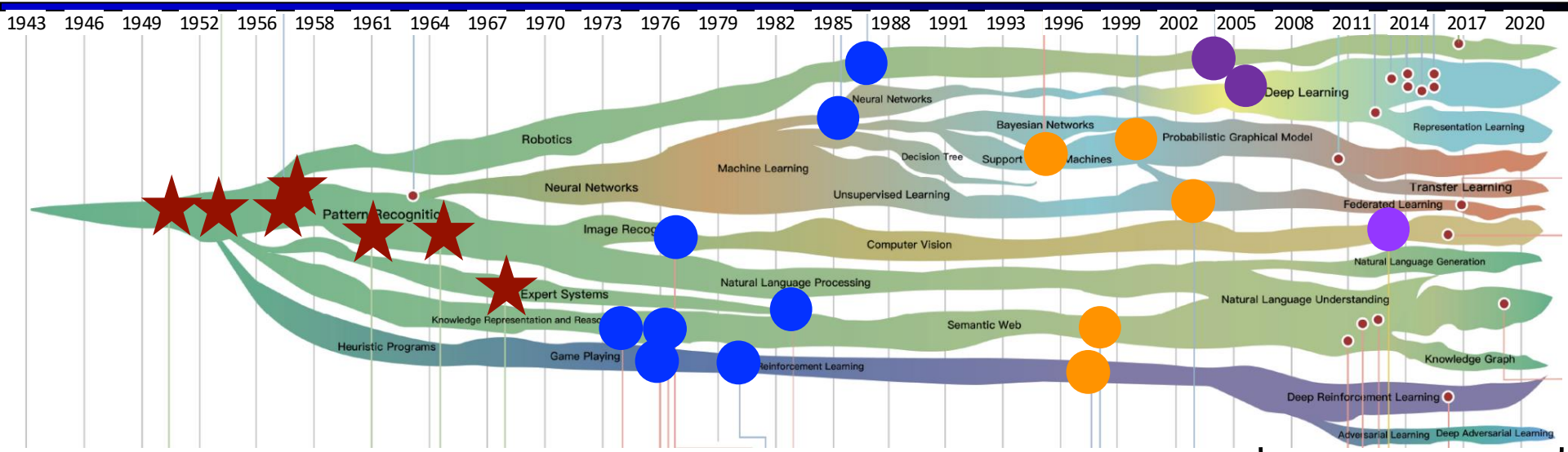
A (Short) History of AI



The Third Wave

- Deep Learning
- Boston Dynamics

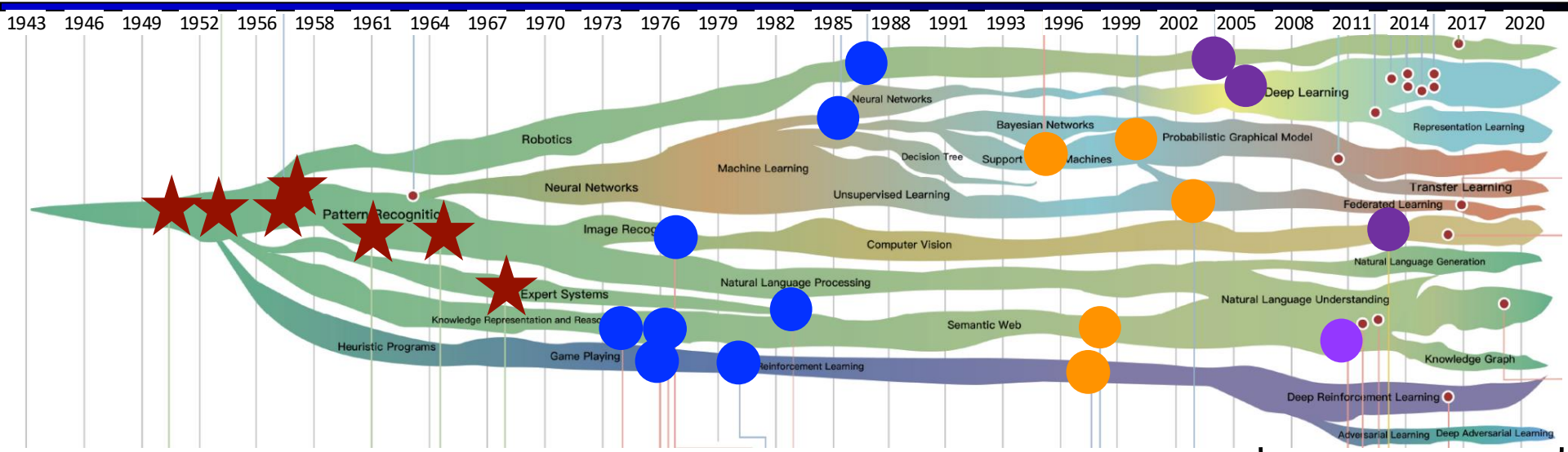
A (Short) History of AI



The Third Wave

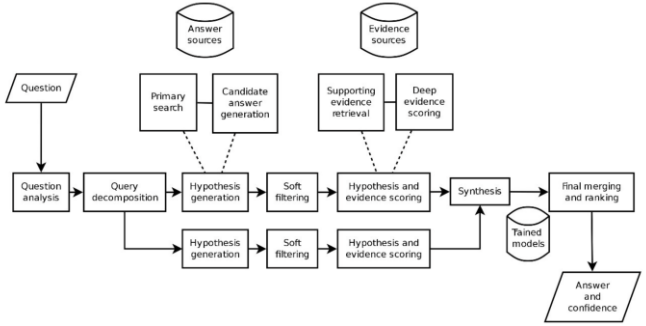
Google self-driving car

A (Short) History of AI

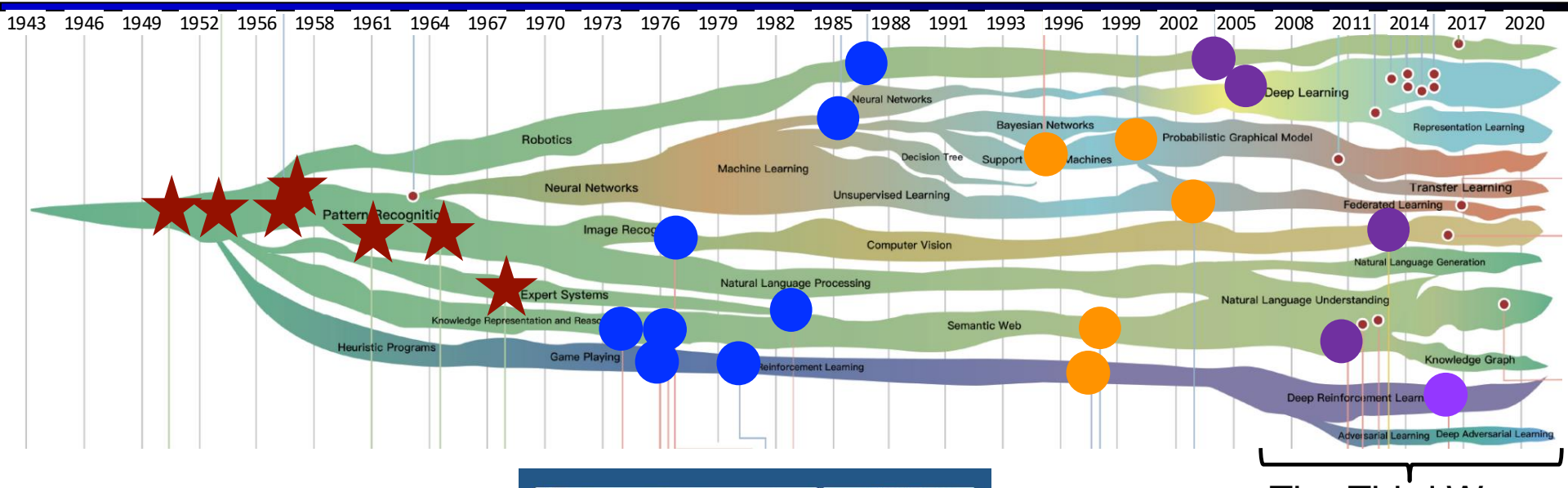


The Third Wave

IBM Watson



A (Short) History of AI



The Third Wave

DeepMind AlphaGo

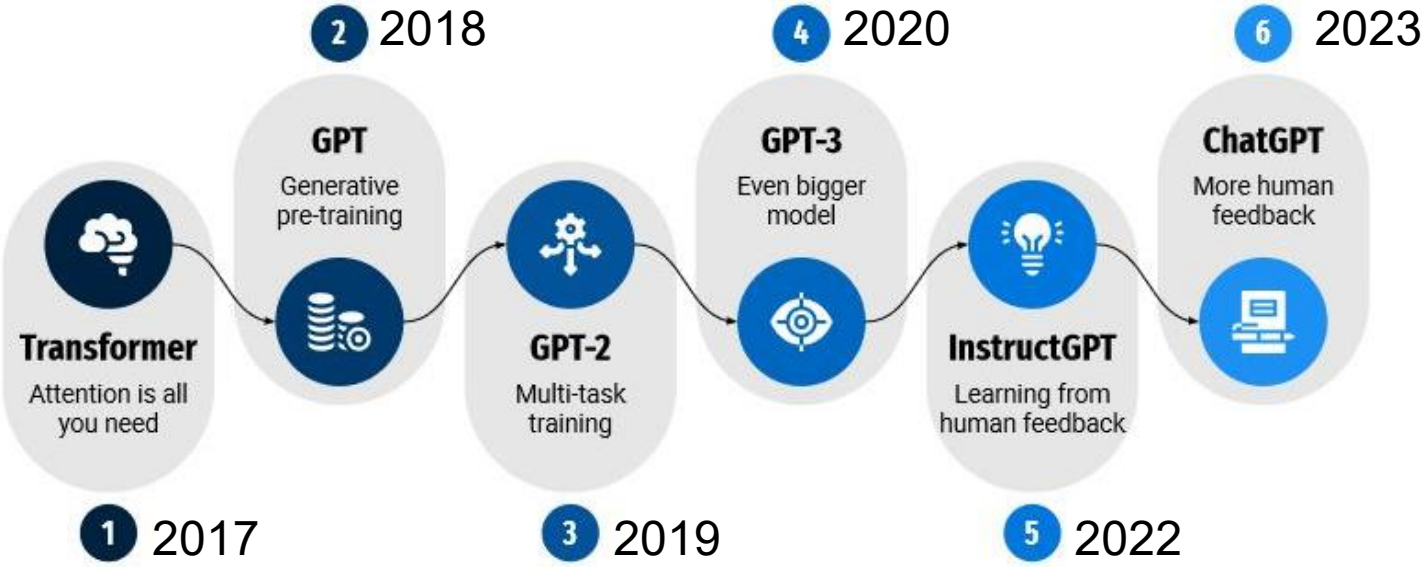
A (Short) History of AI

- What happened after 2020?

The Fourth Wave: Generative & Agentic AI

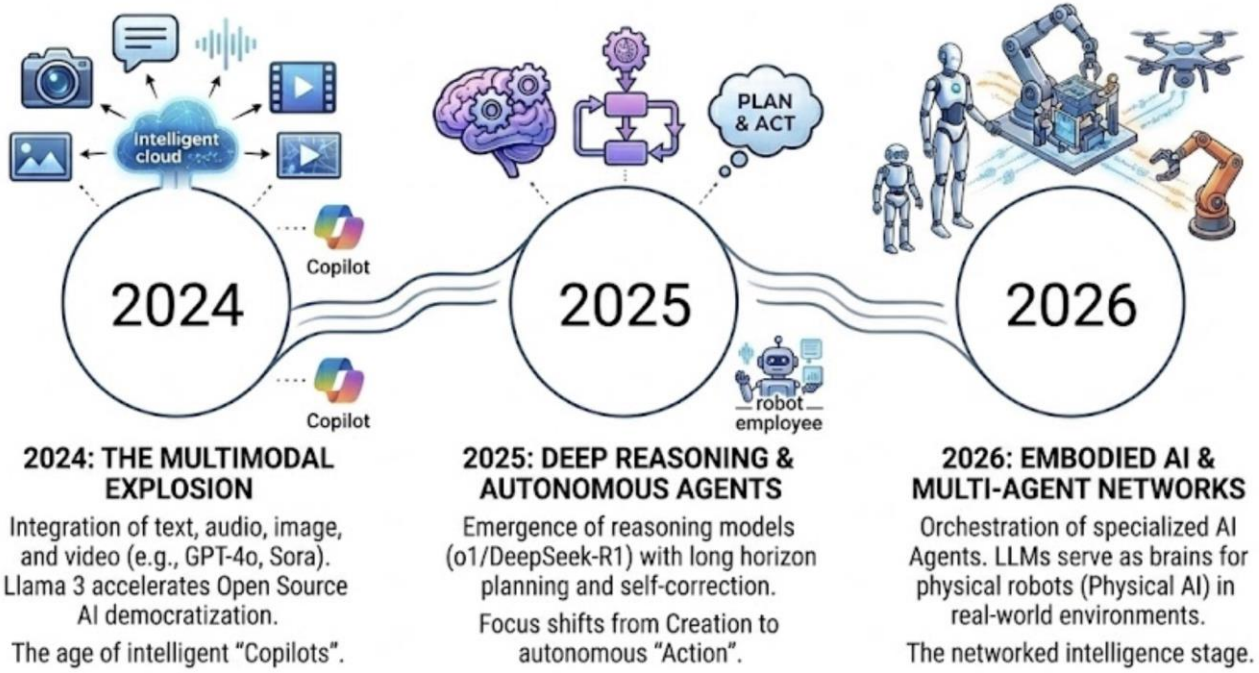
A (Short) History of AI

- What happened after 2020?

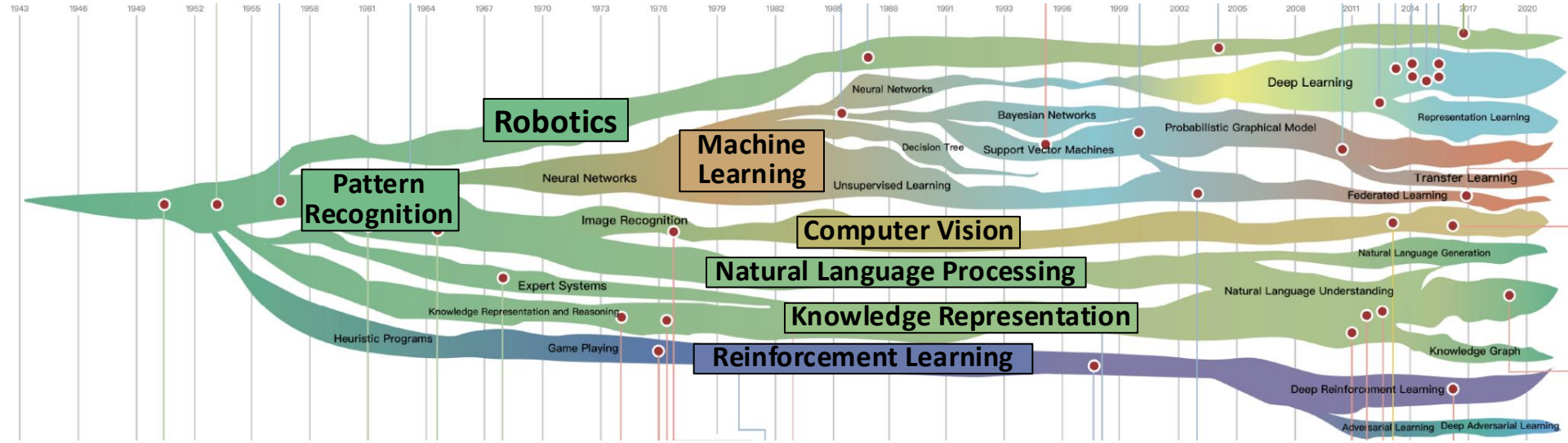


A (Short) History of AI

- What happened after 2020?



What is AI?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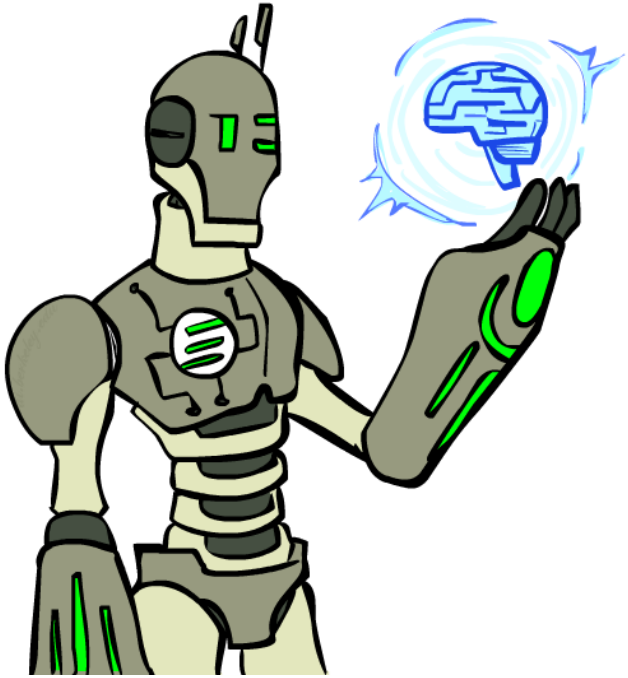
Robotics
Pattern Recognition
Machine Learning
Computer Vision
Natural Language Processing
Knowledge Representation
Reinforcement Learning
Neural Networks
Image Recognition
Expert Systems
Knowledge Representation and Reasoning
Heuristic Programs
Game Playing
Neural Networks
Bayesian Networks
Decision Tree
Support Vector Machines
Unsupervised Learning
Probabilistic Graphical Model
Deep Learning
Representation Learning
Transfer Learning
Federated Learning
Natural Language Generation
Natural Language Understanding
Knowledge Graph
Deep Reinforcement Learning
Adversarial Learning
Deep Adversarial Learning

Today

- What is artificial intelligence?
- **What can AI do TODAY?**
- What is this course?



What Can AI Do TODAY?

- Can AI act as a fully autonomous research assistant or senior programmer, completing complex workflows independently?
  - The evolution of Large Language Models (LLMs) from "Next-Token Prediction" to Autonomous Agents equipped with **Planning**, **Memory**, and **Tool Use**.

Diagnosis-in-the-Loop: Targeted Adversarial Generation via LLM-Based Failure Analysis

Author Name
Affiliation
email@example.com

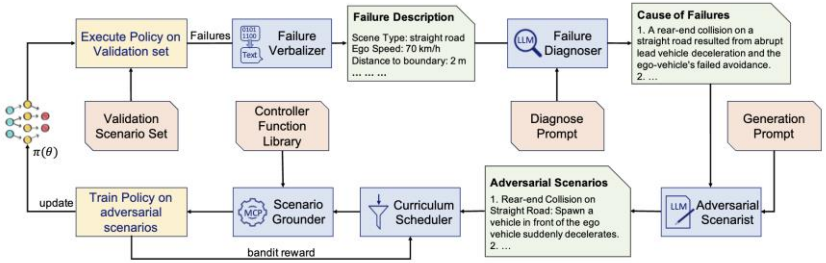


Figure 1: Overview of DiAL. The framework closes the loop between analysis and generation via a four-stage pipeline: (1) The Failure Diagnoser pinpoints semantic root causes from failure trajectories; (2) The Adversarial Scenarist aggregates insights to synthesize targeted natural-language recipes; (3) The Scenario Grounder translates recipes into executable simulations via a Model Context Protocol (MCP); and (4) The adaptive Curriculum Scheduler dynamically schedules scenarios based on agent competency. This cycle transforms passive explanations into active, targeted optimization signals.

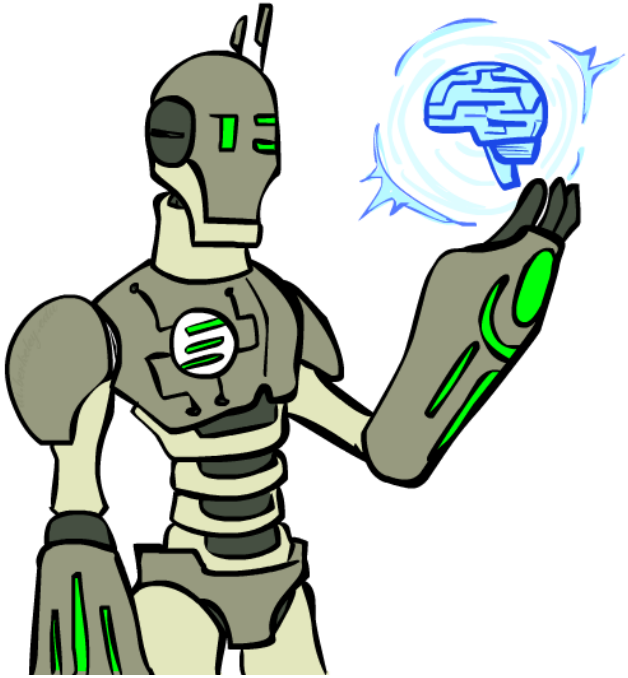
Crawl top-tier papers on **Imitation Learning** from the past decade, extract core innovations, and plot a trend graph using Python.

What Can AI Do TODAY?

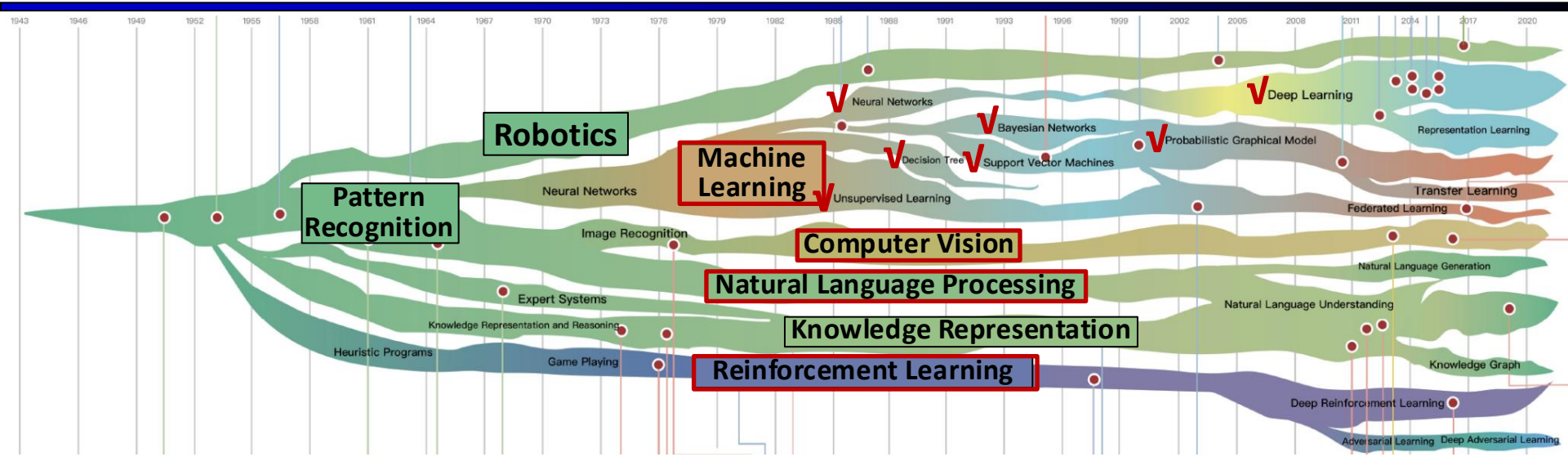
- Drawing bounding boxes to track pedestrians is outdated.
- Can AI truly understand and simulate the physical laws of our world?

Today

- What is artificial intelligence?
- What can AI do?
- **What is this course?**



What is this course?

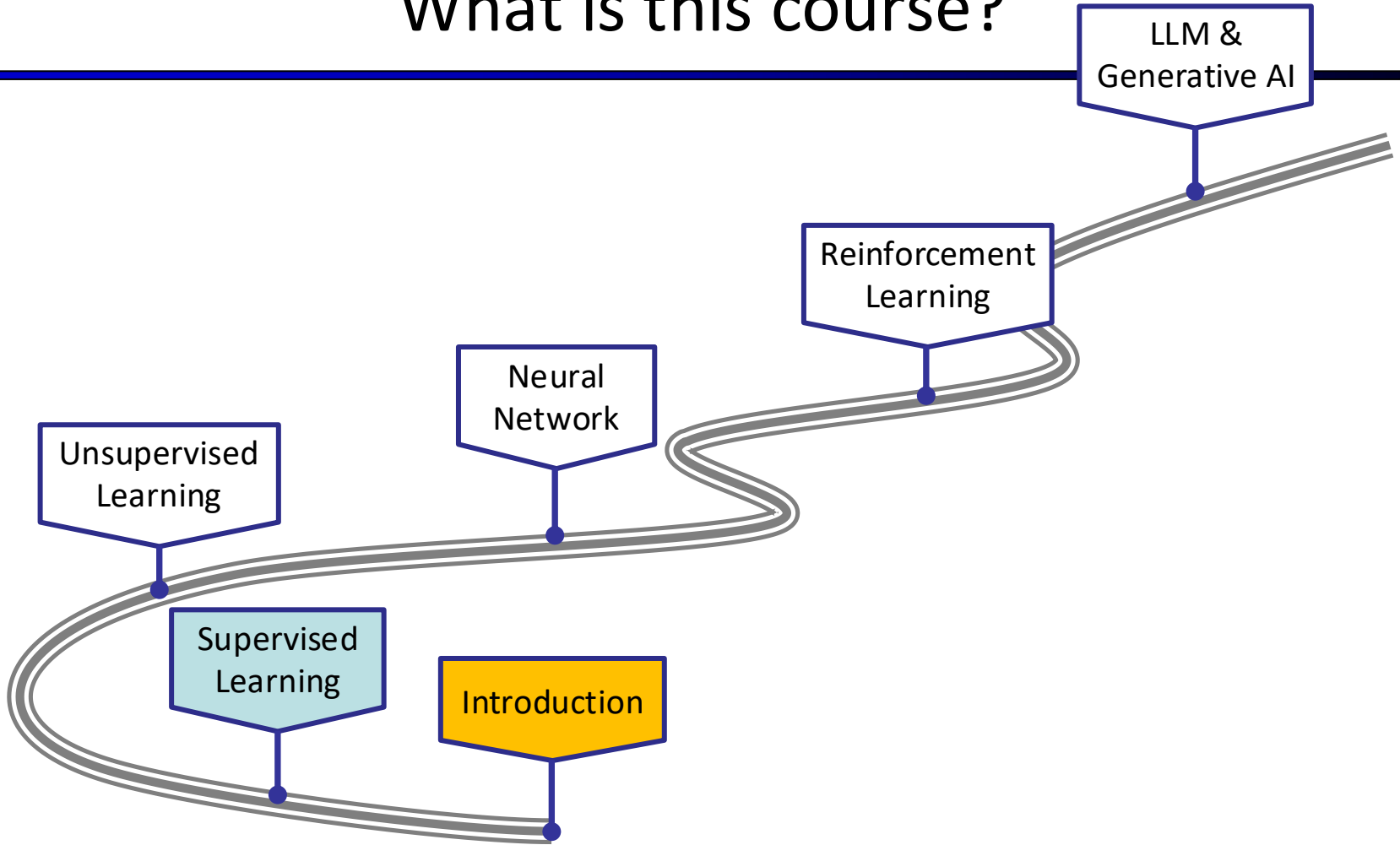


Machine Learning

Reinforcement Learning

Computer Vision

Natural Language Processing

- ✓ Supervised Learning
- ✓ Unsupervised Learning
- ✓ Neural Network

What is this course?